☑ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. **Project No.:** WW-4705R
Project Name: Hilo WWTP Phase 1 **Submittal Number:** 11555-001.1
Submittal Title: Resubmittal Tekleen Product Data ☐ For Information Only
TO: County
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier		
Spec: 11555	**Paragraph:** 1.04-B & 1.04-C	
Authored By: Miguel Serano		**Date Submitted:** 09/12/2025

Submittal Certification		
Check Either (A) or (B):		
☐ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with no exceptions.	
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings except for the deviations listed.	
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.		
General Contractor's Reviewer's Signature:	[signature]	
Printed Name and Title:	Matthew Chun, QC Engineer	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.		
Firm:	**Signature:**	**Date Returned:**

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY THIS CONTRACTOR. IT IS CERTIFIED CORRECT, COMPLETE, AND IN COMPLIANCE WITH CONTRACT DRAWINGS AND SPECIFICATIONS. ALL AFFECTED CONTRACTORS AND SUPPLIERS ARE AWARE OF, AND WILL INTEGRATE THIS SUBMITTAL (UPON APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/12/2025
SPECIFICATION SECTION # 11555
SPECIFICATION Self Cleaning Strainers
PARAGRAPH 1.04
DRAWING n/a
SUBCONTRACTOR n/a
SUPPLIER n/a
MANUFACTURER n/a
CERTIFIED BY CQCM or Designee : [signature] Mark McCarthy

Review Comments:

1. Shop drawings are missing equipment tags. Provide project designated equipment tag numbers as required per specification section 01330 – 1.08.B..

   Incorporated -[Tekleen] will make sure the filter is tagged with the 09-STR-1300.

2. Provide a NEMA 4 enclosure for the controller as required per specification section 11555 – 2.05.E.2..

   Incorportated [Tekleen] will provide the controller with NEMA 4 enclosure. updated SO030387. See CAB PC 303018 T for the enclosure spec sheet.

3. Provide wiring diagram that shows wiring terminals for Running and Failed PLC discrete input points per contract drawing 09-N-01-803 and Section 11555 – 2.05. E. 6.

   Incorporated - [Tekleen] sent over the updated SO030387. See WIRE-GB6-DRYCONTACTS-03 – Backwash & Alarm

4. Provide control station start and stop buttons, an indicating cleaning light, and indicating high differential pressure light per Section 11555 – 2.05. E. 2.

   Incorporated - See GB6-110VAC-NEMA4-CUT-01, GB6 controller inside NEMA4 enclosure 1 & GB6 controller inside NEMA4 enclosure 2.

5. Provide catalog cutsheets on inlet/outlet pressure gauges as provided under Section 11555 – 2.06. A.1.

   Incorporated - See Gauge_Bourdon_AH

6. Confirm spare parts will be provided per Section 11555 – 2.08.A.

   Incorporated - See item #5 from updated SO030387

Submittal Review Response

Project Name:	***Hilo WWTP Rehabilitation and Replacement Project Phase 1***
Submittal No.:	***11555-001.0***
Date:	***5/22/2025***

Client:	County of Hawaiʻi	**Carollo Project No.:**	203975
Contractor:	Nan, Inc.		
Submittal Name:	Tekleen		
Reviewed By:	Francisco Martinez, John Lin		

SUBMITTAL REVIEW		
Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.		
Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☒	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

1. Shop drawings are missing equipment tags. Provide project designated equipment tag numbers as required per specification section 01330 – 1.08.B..
2. Provide a NEMA 4 enclosure for the controller as required per specification section 11555 – 2.05.E.2..
3. Provide wiring diagram that shows wiring terminals for Running and Failed PLC discrete input points per contract drawing 09-N-01-803 and Section 11555 – 2.05. E. 6.
4. Provide control station start and stop buttons, an indicating cleaning light, and indicating high differential pressure light per Section 11555 – 2.05. E. 2.
5. Provide catalog cutsheets on inlet/outlet pressure gauges as provided under Section 11555 – 2.06. A.1.
6. Confirm spare parts will be provided per Section 11555 – 2.08.A.

CAB PC
303018 T

CAB CAB PC 303018 T

Sample photo

Order symbol:	CAB PC 303018 T	**EAN:**	6418074020130
Product nbr:	8113040	**El. number:**	
Description:	Cabinet, PC	Finland:	3424157
Remarks:	Latches on the long side	Denmark:	8212020236
		Italy:	7822025
		Sweden:	2539941
		Norway:	

Including: Cabinet base, door with EPDM gasket, latches, screws for mounting plate/DIN-rail and corner plugs.

Dimensions:	Length	Width	Height
mm:	300	300	180
inch:	11.8	11.8	7.1

Materials:

Material:	Polycarbonate
Base colour:	RAL 7035
Cover colour:	Smoked grey
Gasket material:	EPDM

Temperatures:

Short term:	-40 … 120 °C
Continuous:	-40 … 80 °C
Short term:	-40 … 250 °F
Continuous:	-40 … 175 °F

Accessories:

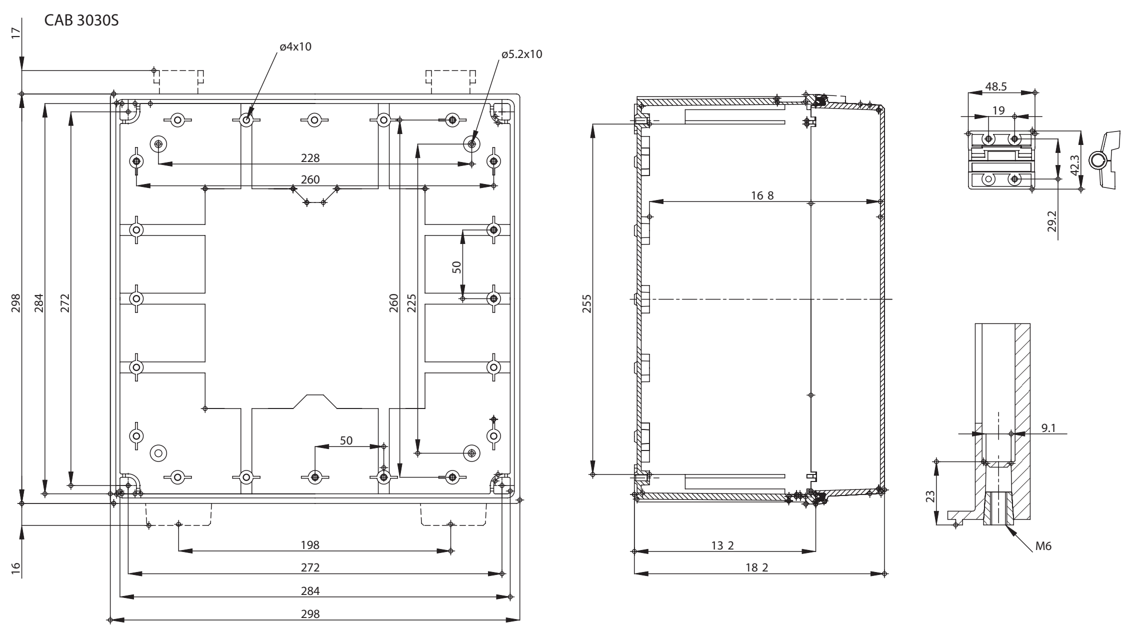
EKIV 33	Mounting plate
EKIV 27	DIN-35 Mounting rail
DS35270E1	DIN-35 Mounting rail
FP A 33	Swing-out door/Front plate
MB 12225 AL	Swing door set
FP 33	Front plate
FP 12-33	Front plate
FP 12x2-33	Front plate
DRF 33	DIN rail frame set

Rating:

Ingress Protection (EN 60529): IP 65
Impact Resistance (EN 62262): IK 08
Electrical insulation: Totally insulated
Halogen free (DIN/VDE 0472, Part 815): Yes
UV resistance: UL 508
Flammability Rating (UL 746 C 5): UL 746C 5V
Glow Wire Test (IEC 695-2-1) °C: 960
UL Type: UL TYPE 1, 4, 4X, 12, 13

Certificates:

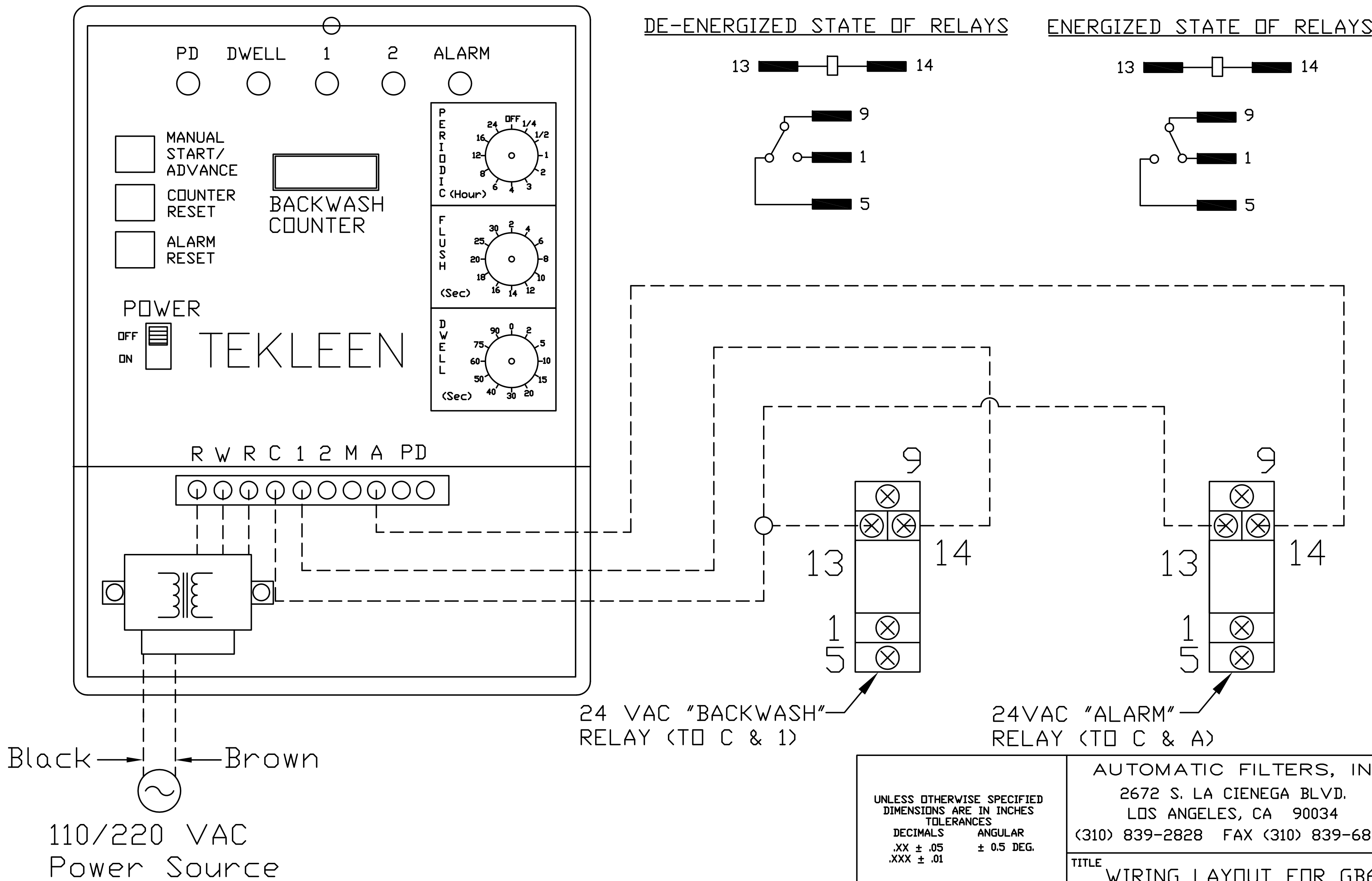
Germanischer Lloyd (GL)
Underwriters Laboratories
Fimko
Lloyds Register of Shipping (LR)
Europe EN 62208: 2003
UL Canada
Gost R



WIRE-GB6-DRYCONTACTS-03 – Backwash & Alarm

GB6 ELECTRIC CONTROLLER

REVISIONS				
ZONE	REV	DESCRIPTION	DATE	APPROVED



NOTES:
1. WHEN THE BACKWASH CYCLE AND THE ALARM ARE TRIGGERED, TERMINALS 5 & 9 OPEN CREATING NO VOLTAGE BETWEEN THOSE TWO TERMINALS ON THE RELAYS RESPECTIVELY.

Junction	○
No Junction	⌒
Electric Connection	----

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES

DECIMALS	ANGULAR
.XX ± .05 .XXX ± .01	± 0.5 DEG.

DRAWN: S. PERKINSON	DATE: 05/17/13
CHECKED	
DESIGN	

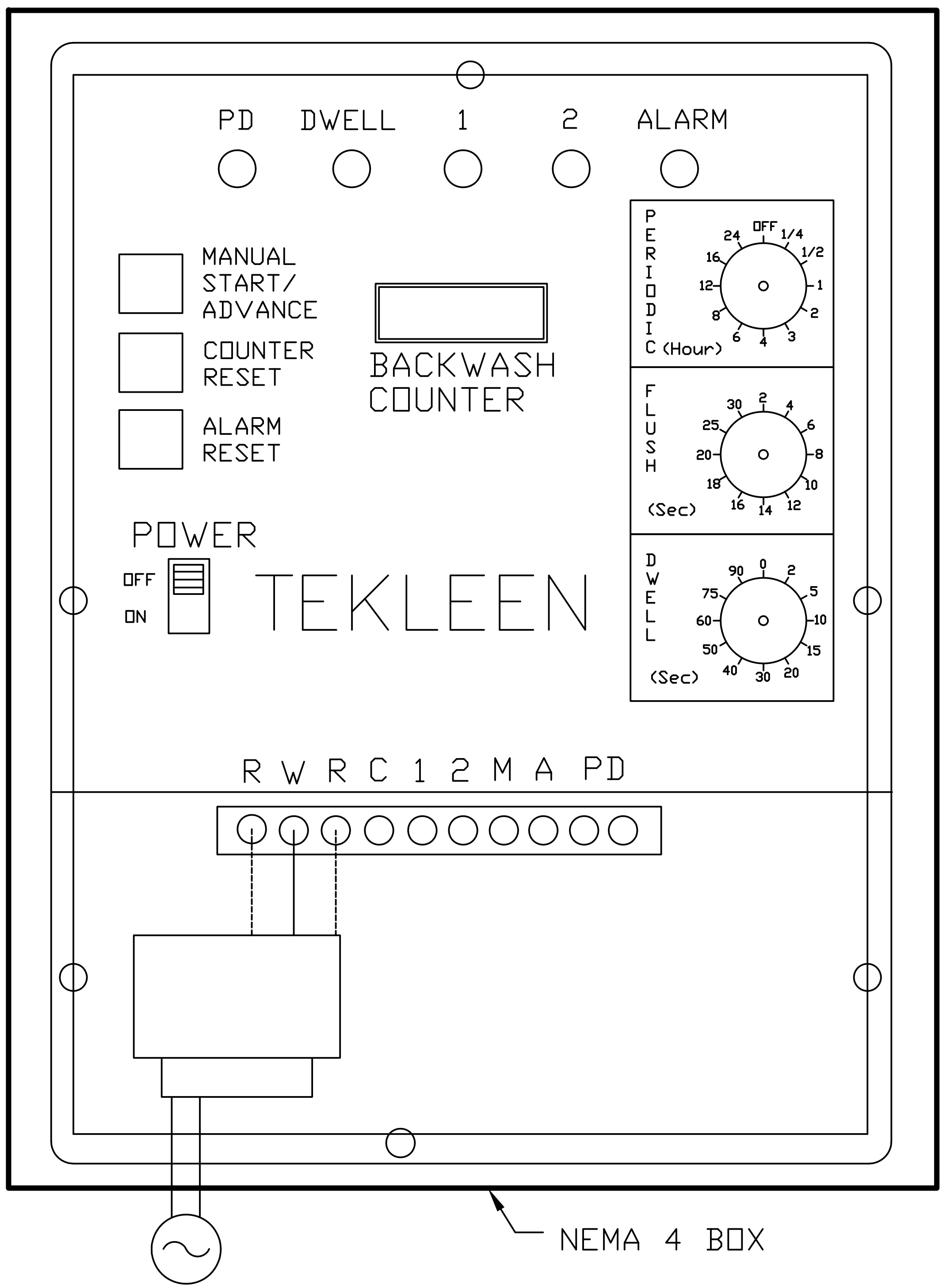
AUTOMATIC FILTERS, INC.
2672 S. LA CIENEGA BLVD.
LOS ANGELES, CA 90034
(310) 839-2828 FAX (310) 839-6878

TITLE: WIRING LAYOUT FOR GB6 CONTROLLER WITH DRY CONTACTS FOR BACKWASH & ALARM

SIZE	DWG NO.	REV
A	WIRE-GB6-DRYCONTACTS-01	N/A
NOT TO SCALE		SHEET 1 OF 1

GB6-110VAC-NEMA4-CUT-01

GB6 ELECTRIC CONTROLLER



REVISIONS

REV	ZONE	DESCRIPTION	DATE	APPROVED

Available Features

LCD Monitor Panel
Backwash Counter
Periodic Flush Timer
Manual Backwash
Weather-proof, UV Resistant
Protective Case
NEMA 4 Enclosure

Specifications

Input: 110 VAC, 60 Hz.
Output: 2.5 amps at 24 VAC
Efficiency: Less than 20 milliamps (idle)
Dimensions (Inside Box): 10″L X 7.5″W X 4″H
Dimensions (NEMA 4 Box): 11.8″L X 11.8″W X 7.1″H
Weight: ~12 lbs.

Terminal Connections

C: Common
1: Filter 1 Control
2: Filter 2 Control
M: Main Line (PSV)
A: Alarm
P/D: Differential Pressure

NOTES:
1. OVERALL DIMENSIONS MAY VARY SLIGHTLY DEPENDING ON IP56/NEMA 4 BOX USED.

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES

DECIMALS	ANGULAR
.XX ± .05 .XXX ± .01	± 0.5 DEG.

AUTOMATIC FILTERS, INC.
2672 S. LA CIENEGA BLVD.
LOS ANGELES, CA 90034
(310) 839-2828 FAX (310) 839-6878

DRAWN: J. KAWAMOTO	DATE: 07/23/13	TITLE: GB6 ELECTRIC CONTROLLER IN NEMA 4 ENCLOSURE, 110 VAC	
CHECKED		SIZE: A	DWG NO.: GB6-110VAC-NEMA4-CUT-01 / REV: N/A
DESIGN FOR:		NOT TO SCALE	SHEET 1 OF 1

See Gauge_Bourdon_AH

AH

Polypropylene Case Pressure Gauge (4 ½'')
Bourdon type

GOST - R
IP 65

GOST

Special Features

- Polypropylene case with SS measuring system
- Wetted parts AISI 316L stainless steel or Brass
- Accuracy ± 0.5% F.S.
- Solid front design features baffle wall interposed between the sensing system, the window and a pressure relieving back for increased safety
- Standard followed ASME - B 40.100

Application

- Suitable for high pressure gas
- Corrosive / hazardous environment that will not obstruct the pressure system.

Specifications

Standard Version : 4½"

Accuracy	:	± 0.5% F. S. (As per ASME B40.100)
Ambient temperature	:	- 20°C to + 65°C
Process temperature	:	Max 150°C
Operating pressure range	:	75 % of Scale Value
Over pressure limit	:	≤ 100 bar : 125% of Max. Scale Value
	:	> 100 to ≤ 600 bar : 115% of Max. Scale Value
	:	> 600 to ≤ 1600 bar : 110% of Max. Scale Value

Case	:	Fibre Glass Reinforced Polypropylene
Bourdon	:	AISI 316L SS or Brass
Socket	:	AISI 316L SS or Brass
Movement	:	AISI 304 SS
Joints	:	Tig Argon Arc Welding

Protection	:	IP 65
Dial	:	Aluminum, black graduation on white background
Pointer	:	Aluminum, black Colored, Micrometer Zero Adjustable
Window	:	Plexiglas
Blow off Disc	:	Fibre Glass Reinforced Polypropylene
Gasket	:	Compatible with glycerin or silicone

Dry But Fillable Version (Standard)

Fillable Dampening Liquid	:	Glycerine 99.7%
Ambient Temperature	:	Maximum 65°C
Process Temperature	:	Maximum 65°C
Other Features	:	Refer Specification of Standard Version

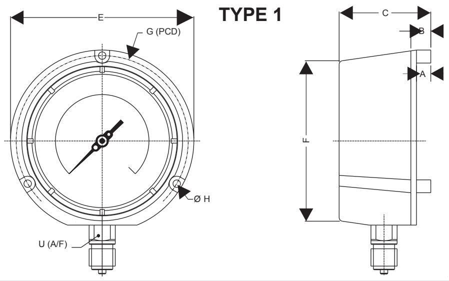
Temperature effect:
The variation of indication caused by effects of temperature is to be calculated by below formula; which is to be added in the specified accuracy while measurement :- Formula : ± 0.04 x (t2 - t1) % of F. S. where t1 = reference temperature (+20°C) and t2 = ambient temperature in °C.

AH

Polypropylene Case Pressure Gauge (4 ½'')
Bourdon type

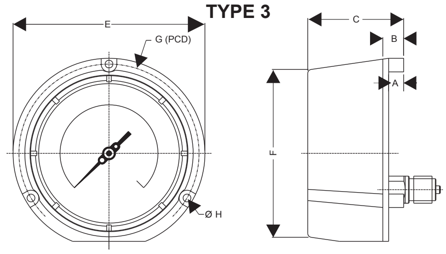
Dimensions - standard version

TYPE 1



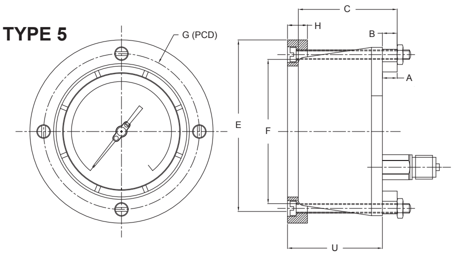
NS	A	B	C	E	F	G	H	U	Weight in gram (With Box)	Weight in gram (With Glycerin & Box)
4½"	12.5	24.5	73.5	148	130	137.5	6	22	900.0	1300.00

TYPE 3



NS	A	B	C	E	F	G	H	U	Weight in gram (With Box)
4½"	12.5	24.5	73.5	148	130	137.5	6	22	850.0

TYPE 5



NS	A	B	C	E	F	G	H	U	Weight in gram (With Box)
4½"	12.5	24.5	73.5	148	130	137.5	6	22	1000.0

Notes : • Drawings are not to scale. • All Dimensions are in mm. • NS = Nominal Size • H - Arrangement for Mounting type 1 • Weights mentioned are approximate and for standard product. • Weight can be different after selection of options.

Note : We offer National / International Scales like kPa, MPa, bar, psi, kg/cm² & Dual Scale like kPa with psi, kPa with bar, bar with psi or Equivalent scales as per the requirement can be provided on request. Following are the example tables for kg/cm² & psi scales

Pressure ranges

Code	Bar	Code	kPa	Code	kg/cm²	Code	psi
C59	-1 ... 0 [3]	D59	-100 ... 0 [3]	F59	-1 ... 0 [3]	H59	-30″ Hg ... 0 [3]
C72	-1 ... 0.6	D72	-100 ... 60	F72	-1 ... 0.6	H73	-30″ Hg ... 15
C74	-1 ... 1.5	D74	-100 ... 150	F74	-1 ... 1.5	H75	-30″ Hg ... 30
C76	-1 ... 3	D76	-100 ... 300	F76	-1 ... 3	H2C	-30″ Hg ... 60
C77	-1 ... 5	D77	-100 ... 500	F77	-1 ... 5	H78	-30″ Hg ... 100
C79	-1 ... 9	D79	-100 ... 900	F79	-1 ... 9	H79	-30″ Hg ... 150
C81	-1 ... 15	D81	-100 ... 1 500	F81	-1 ... 15	H81	-30″ Hg ... 220
C82	-1 ... 24	D82	-100 ... 2 400	F82	-1 ... 24	H82	-30″ Hg ... 300
C12	0 ... 0.6 [2]	D12	0 ... 60 [2]	F12	0 ... 0.6 [2]	H13	0 ... 10 [2]
C15	0 ... 1 [3]	D15	0 ... 100 [3]	F15	0 ... 1 [3]	H15	0 ... 15 [3]
C16	0 ... 1.6	D16	0 ... 160	F16	0 ... 1.6	H1C	0 ... 20
C18	0 ... 2.5	D18	0 ... 250	F18	0 ... 2.5	H17	0 ... 30
C19	0 ... 4	D19	0 ... 400	F19	0 ... 4	H19	0 ... 60
C20	0 ... 6	D20	0 ... 600	F20	0 ... 6	H21	0 ... 100
C22	0 ... 10	D22	0 ... 1 000	F22	0 ... 10	H22	0 ... 160
C24	0 ... 16	D24	0 ... 1 600	F24	0 ... 16	H23	0 ... 200
C26	0 ... 25	D26	0 ... 2 500	F26	0 ... 25	H25	0 ... 300
C27	0 ... 40	D27	0 ... 4 000	F27	0 ... 40	H26	0 ... 400
C29	0 ... 60	D29	0 ... 6 000	F29	0 ... 60	H27	0 ... 600
C31	0 ... 100	D31	0 ... 10 000	F31	0 ... 100	H30	0 ... 1 000
C33	0 ... 160	D33	0 ... 16 000	F33	0 ... 160	H31	0 ... 1 500
C35	0 ... 250	D35	0 ... 25 000	F35	0 ... 250	H34	0 ... 3 000
C38	0 ... 400	D38	0 ... 40 000	F38	0 ... 400	H38	0 ... 6 000
C39	0 ... 600	D39	0 ... 60 000	F39	0 ... 600	H40	0 ... 10 000
C41	0 ... 1 000 [1]	D41	0 ... 100 000 [1]	F41	0 ... 1 000 [1]	H41	0 ... 15 000 [1]
C42	0 ... 1 600 [1]	D42	0 ... 160 000 [1]	F42	0 ... 1 600 [1]	H1D	0 ... 20 000 [1]

Bourdon

AH

Polypropylene Case Pressure Gauge (4 ½'')
Bourdon type

AHG6-A50.H22

How To Order			Example
Basic Model			AH
Wetted Material			X
316L Stainless Steel	G		
Brass	A		
Nominal Size	4 ½''		6
Type of Mounting			X
A Wall / surface / projection mounting with bottom entry			
E Direct lower back entry (never fillable)			
B Lower back entry with front flange (never fillable)			
Gauge Connection			X
J 3/8" BSP (M)			
3 ½" BSP (M) Standard			
5 1/4" NPT (M) Standard			
6 ½" NPT (M) Standard			
Liquid filling			
Dry	**0**		
BH2: high viscosity glycerin 99.5% (medium : 0 ... +90°C)	2		
BH1: low viscosity glycerin/water 86% (medium : -20 ... +70°C)	1		
BH3: silicone oil (medium : -40 ... +100°C)	3		
BH4: silicone oil (medium : -60 ... +100°C)	4		
BH5: fluor carbon for oxygene use (160 bar max.) (-15 ... +100°C)	5		
Range			XXX
Refer range table			
Optional Extras			XX

TF	Conformity as per NACE Standard	**SG**	Oxygen service (for dry version)
TI	Monel Bourdon & socket (Monel version)**	**SF**	Movement with dampening jelly
PW	Five point calibration certificate	**QB**	Integral dampening screw (AISI 316 SS)
PU	Calibration certificate traceable to National / International standards	**FL**	Integral dampening screw (monel)**
RH	Custom designed dial	**RX**	Shatterproof/safety glass

AHG6-A50.H22

Ordering example with options

AHG	6	_	A	3	0	.	C22	/		_		_

- Polypropylene / Stainless steel
- Nominal size 130 mm
- Bottom connection, 3 back lugs fixing
- Process connection G1/2
- No liquid filling
- Scale bar : 0 ... 10 bar
- Option : Anti-vibration dampening movement
- Option : Oxygen application
- Option : Restrictor screw Ø 0.5

GB6 controller
inside NEMA4
enclosure
1


PD DWELL 1 2 ALARM
MANUAL START/ ADVANCE
COUNTER RESET
ALARM RESET
BACKWASH COUNTER
PERIODIC (HOURS)
OFF 1/4 1/2 1 2 3 4 6 8 12 18 24
FLUSH (SEC)
2 4 6 8 10 12 14 16 18 20 25 30
DWELL (SEC)
0 2 5 10 15 20 30 40 50 60 75 90
POWER
OFF
ON
TEKLEEN
Automatic Filters Inc.
(310) 839-2828 U.S.A.
MODEL GB6/GB7
OUTPUT FUSE 3 AMPS
12VDC → + −
24VAC → R W R C 1 2 M A PD
info@tekleen.com
www.tekleen.com

GB6 controller inside NEMA4 enclosure 2.


TEKLEEN®
Automatic
PD DWELL 1 2 ALARM
MANUAL START/ ADVANCE
COUNTER RESET
ALARM RESET
BACKWASH COUNTER
PERIODIC
(HOURS)
FLUSH
(SEC)
DWELL
(SEC)
POWER
OFF
ON
TEKLEEN
Automatic Filters Inc.
(310) 839-2828 U.S.A.
MODEL GB6/GB7
OUTPUT FUSE 3 AMPS
12VDC → +
24VAC → R
– W R C 1 M A PD
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